

Deep Pump

The deep buoyancy pump (350 m and 1000 m, ~900 cc) is a **rotary displacement (hydraulic)** design. Instead of moving seawater, it moves **oil** from an **internal reservoir** to an **external bladder** to change the vehicle's buoyancy. A rotary valve controls the flow of oil from the bladder back into the reservoir.

Source

Paraphrased and consolidated from the *Slocum G3 Glider Operators Manual*, the TWR user forum, and the UG2 community Slack. Pumps are **factory configured** — contact glidersupport@teledyne.com before changing pump settings. See also the [Shallow Pump](#).

Vacuum requirement

Operate under vacuum

The glider must be **under vacuum** while running the deep pump — the internal vacuum is what draws oil from the external bladder back into the body. Running it without vacuum won't *damage* the pump, but the oil won't retract properly.

The deep pump can **retract at any depth** (unlike the shallow pump, which is limited to its rated range). Even so, deep gliders are rarely inflected at the top for sustained operation below ~75 m.

The G3 HD pump & 3-way valve

The deep pump was redesigned for the G3 to be more robust and easier to maintain. It uses the device name `hd_pump` (must be in the installed-devices list in `autoexec.mi`) and adds a **3-way ball valve** controlling oil flow from the external bladder back into the reservoir.

- In normal operation the valve **opens at the surface** to retract oil and begin the dive, then stays **closed for the whole dive** (including the bottom inflection,